

# Step	Section Title	Instructions	Tools	Materials
1	Foot Plates	Measurements from installation floorplan, Laser lines/string lines to align footplates, check diagonal squariness. , mark for exchange table but dont install footplates	Laser line, String, Concrete Drill,16mm concrete drill, Battery drill, impact driver and sockets	m16 x100 screwbolt [x40], Foot plates [x20]
2	Feet Into Body	120mm gap between base and ground clearance for slag collection trays - Welded to footplates after machine leveled	Overhead Gantry/Firana hoist, large spanners, stick welder	Welding rods
3	Connect Exchange Table Body	Connect exchange table body - Connect with threaded rod, not placed on foot plates initially - placed on roller wheels for ability to pull away for access to chain drive motor install and adjustment. rods assist with ease on alignment later. foot plates can be installed with table in location later and welded after leveling	Roller wheels [x4]	m16 Threaded rod [2m/0.5mx4off]
4	Bed Leveling	Use free linear bearing & mag base dial gauge to run along rack to ensure straightness. - when unpackaging racks be careful not to remove lettering hints as the racks have mbeen drilled in place and have designated location on machine base. Bed machined surfaces will require deburring and cleaning and potentially thread mounting holes tapped to clean swarf from manufacturing	Laser level with tripod and reciever, Bubble level, Deburr stone, Files, Torque wrench, Long level / straight edge, Scouring pads, Bubble levels, Dial gauge with mag base, Angle grinder with fine sanding disk, deburr knives, taps for m5, 6, drill bit selection	Slideway oil, WD40, Cotton Rags
5	Chain & Table Install	Chain, chain drive motor and chain clips installation. Install table with removed slats so we can set chain tension prior to body panels install. After chain tensioned exchange table frame can be pushed toward main body for foot plate installation and moving out exchange table for tables wheel alignment. will require forklift and straps for moving of exchange tables.	Hammer[Deadblow and standard], Forklift[ideally lifting boom attachment], Spanner, allenkeys and socket sets, string line, Lifting straps and shackles	Bubble level, deburr stone, slideway oil, files, torque wrench, long level / straight edge. Scouring pads, WD40, anti-seize
6	Linear Rails & Cross Beam	Ensure linear bearing and underside of crossbeam gantry are deburred and clean before assembling - Connect oil lines tidily before placing on linear bearings. Install cross beam with bolts only finger tight to linear bearings.	Rags, Whetstone / Deburring tools	Included in kit
7	Squaring Machine	Place square in centre of machine on sturdy table - leveled with packers, Mount dial gauge to laser head mounting screw holes with adapter bolt. Manually move crossbeam along Y axis to align square and clamp in position. Push gantry along X axis and adjust crossbeam with deadblow until x and y axis are 0.2mm/m straight. If awkward: continue cables and installation of required electrics before squaring of crossbeam. After squaring tighten crossbeam mounting bolts and recheck. recheck at a different location on machine.	Sturdy table, Packers, Dial gauge, Adapter Bolt, Square, Clamps, Torque Wrench, Deadblow	N/A
8	Motors & Drive Gear	Install Y axis motor mount bolts at finger tight, push drive gear into racks untill fully bound with no play. Then tighten mounting bolts 30 degrees past hand tight, then do same to tension bolts so some preload is on driving gear.	Spanner set, socket set with extensions swivel joints	Included in kit
9	Front Body panels and Valve Island	Install cable trays and front body panel for valve island (Extraction door numatics) connections.		Included in kit
11	Running Cables, and gas lines and installing cable tray	Install side cable trays, oil reservoir, gas regulators, run gas lines, power lines	Electrical Toolkit	Oil, Gas Fittings [], Gas hoses [], Sealants/Glue []

# Step	Section Title	Instructions	Tools	Materials
10	Panels & Wall Frame	Install front panels and rear pillar bodys without tightning mounting bolts for adjustment when installing wall panels. Install side wall framing then remove and place on layed out wall panels. drilling and taping of holes will be neccisary for proper panel alignment. after wall sections align well. remove centre wall panels for wall frame mounting bolt access and lift into location with forklift or rmanpower. Adjust front and back pillars as necessary.	Clamps, Drills, Laser line, Forklift, Straight edge, Drawn out measurements of wall assemble width tolerance.	Large Cardboard roll or Plywood Panels for placing wall panels safely on ground. Large Clear area
11	Electrics and sensors, contol panels gas hookup, additional	Note before reinstalling oil reservoir and installed wall gas lines, power lines panels, should be installation of table and gas lines - locking system attach side panels and install and laser head. Fibre cable install wire drive chain motor a back body frame and panels next at a install covers.	Wire routers	Oil reservoir, Gas lines, Power lines, Side panels, Laser head, Fibre cable, Covers
12	window install	Aluminium window strips will benifit from retapping at large diameter since manufacturing quality is low, drilling of clearence holes in machine wall panels will also be required. loosely install bottom alu rail and remove bolts on top alu rail so window glass is seated in bottom rail and top rail can be pushed in and second installer inside machine can tighten rails to hold window in place.	Window Suction cups, m5 tap and 6mm drill	Loctite